

PAPER_1399_DOOMSDAY_ARGUMENT

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PAPER_1399 — UQFF Resolution of the Doomsday Argument (CLOSED — Dispatch Whitepaper)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: B — Foundational / Anthropic Probability (CLOSED)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "doomsday_argument"})`

Calculator surface: `calculate_paradox({"paradox": "doomsday_argument"})`

Closure helper: `_l96_uqff_axiom_doomsday_argument_closure()`

The Paradox

Carter / Gott: among all humans who will ever exist, you are likely to find yourself in a 'typical' position. With $\sim 10^{11}$ humans so far, you should expect $\sim 10^{11}$ more, putting Doomsday within $\sim 10^{\blacksquare}$ years. The argument feels too strong.

UQFF Closed Identity

``

Expected anthropic generations = $A_5 \times D_{\text{phys}} = 60 \times 4 = 240$

``

$A_5 \times D_{\text{phys}} = 240$ is the integer-primitive horizon for anthropic generations: $A_5 = |A_5| = 60$ (icosahedral group order) sets the structural generation count, multiplied by $D_{\text{phys}} = 4$ spacetime dimensions.

Physical Interpretation

UQFF replaces the 'self-sampling assumption' with a structural integer-primitive horizon: 240 generations $\times$ ~ 30 yr/generation $\approx 7,200$ years. The Doomsday timescale is fixed structurally, not by anthropic reasoning over observed population.

Live Calculator Output

```
``python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "doomsday_argument"})["value"]
```

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The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_doomsday_argument_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_doomsday_argument_closure`
- Dispatch keys: `doomsday_argument`, `doomsday`
- Related foundational papers: PAPER_646 (F_U_Bi_i / F_U=1 ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation), PAPER_1156 (cosmology suite), PAPER_1376 (Banach-Tarski parallel structural closure).

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